MACHINE GUARDIAN · MAC-ATTACK

Machine Guardian — The Current Model

The bearing-deterioration multi-task MLP selected in Training Run 3: data, inputs, parameters, training, outputs, results and hardware

850
parameters: 8 → 32 → 16 → 2

6 of 7
held-out bearings alarmed before the stop

1.78 µs
per measurement on the 8 × 8 array (RTL simulation)

MAC-Attack senior capstone · Machine Guardian
Configuration E, trained on 7 Paderborn development bearings · 9 November 2026
Source: model/results/supervised/run3/ and docs/Model/10_Training_run3.md
The output is a deterioration score, not a validated probability of physical failure.

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MAC-Attack senior capstone. This document describes only the latest model: the bearing-deterioration multi-task MLP selected in Training Run 3 (configuration E). The earlier models and runs are covered by documents 01–11 in this folder; `README.md` gives the reading order. Every number here comes from `model/results/supervised/run3/` and `docs/Model/10_Training_run3.md`. Written 9 November 2026.

1. The Model in Brief

What it does	Watches one bearing through one accelerometer (plus shaft speed) and raises a **deterioration alarm** when the bearing's vibration looks like a worn-out, end-of-life bearing
Network	multi-task MLP, **8 → 32 → 16 → 2**, ReLU, **850 parameters**
Configuration	**E**: weight decay 0.01, log-compressed inputs, temperature input switched off
Trained on	7 Paderborn run-to-failure bearings: 9,003 measurements, of which 735 are labelled
Output 1	deterioration score (0–1) and a latched alarm. **Trained**
Output 2	Motor Health Index (0–100 %). **Untrained, frozen, never reported**
Held-out result	alarm before the stop on **6 of 7 bearings**; **0** false alarms on healthy rows; never earlier than the simple anomaly detector, which catches all 7
Hardware	INT8 on the team's 8 × 8 systolic array: **382 clock cycles = 1.78 µs** per measurement, bit-exact in RTL simulation. **Not yet run on the ZedBoard**
Status	experimental. Not frozen, and not yet tested on the four locked test bearings

> **Key finding:** the model works end to end, from vibration to an INT8 alarm on our accelerator, and it is stable and reproducible. It does not yet beat the label-free anomaly detector: it alarms at the same time or later, and misses one bearing.

2. The Data

2.1 Datasets and what each is used for

Dataset	Use in the current model
Paderborn run-to-failure bearing dataset (Paderborn University, LDM; Zenodo 10.5281/zenodo.10805042): 17 bearings, each run under varying load and speed until it failed, with two accelerometers, temperature, speed and load	the only training and validation data
— 7 development bearings: B02, B03, B04, B08, B09, B11, B12	training and leave-one-bearing-out validation
— 4 locked test bearings: B06, B07, B13, B17	the final one-time test. **Never loaded**
— B01	processed, but no labels (it overheated within 16 minutes and its documents contradict each other)
— B05, B10, B14, B15, B16	not used yet (candidates for the next run)

2.2 The seven training bearings

Bearing	Sampling	Operating measurements	Recorded span	Stop (log)	Damage found at dismantling
B02	128 kHz	1,065	5.6 h	vibration	inner race
B03	128 kHz	614	3.2 h	temperature	inner race
B04	128 kHz	1,064	5.7 h	vibration	inner race, ball
B08	128 kHz	1,777	10.6 h	temperature	inner race, ball
B09	128 kHz	1,763	12.1 h	vibration	outer race, ball
B11	64 kHz	1,953	8.8 h	vibration	inner race, outer race, ball
B12	64 kHz	767	2.5 h	vibration	inner race, ball
Total		**9,003**	48.5 h		

What a data point is: one 1.6-second vibration recording, taken every 12–23 s. It holds 204,800 samples per channel at 128 kHz, or 102,400 at 64 kHz. Each recording becomes one row of 8 inputs.

2.3 Labels

Label	Rule	Rows	Share of the 9,003
Healthy (0)	first 100 operating measurements of each new bearing	700	7.8 %
End-stage (1), **provisional proxy**	last 5 measurements before a documented stop	35	0.4 %
Unlabelled	everything in between	8,268	91.8 %

- **Source:** the labels come only from the dataset documents (start of test, logged stop, damage found at dismantling). No sensor value is used to set them.
- **What the end-stage labels mean:** they stand for the end-of-test state. When damage actually began is unknown.
- **Health index:** no measured damage sizes exist, so the Motor Health Index has no labels.

3. How the Data Are Split

The split is always by **whole bearing**, never by individual measurement, so a model is always judged on a bearing it has never seen.

Level	Train	Validation	Test
Whole dataset (17 bearings)	13 development bearings (76 %), of which 7 are used	–	4 locked bearings (24 %), untouched
Each outer fold (7 bearings)	6 bearings (86 %)	–	1 held-out bearing (14 %)
Each inner fold, for choosing the configuration	5 bearings (71 % of the 7)	1 bearing (14 %)	the outer bearing (14 %) stays unseen
Final model	all 7 bearings (100 %, 735 labelled rows)	–	–

Nested leave-one-bearing-out:

- **Outer loop:** each of the 7 bearings is held out in turn: 7 folds × 5 seeds = 35 held-out networks.
- **Inner loop:** inside each outer fold, 6 more folds choose the configuration.
- **Final model:** trained on all 7 bearings, so it has no held-out data. Its performance estimate is the 35 held-out networks.
- **Unbiased estimates:** only the four bearings added in Run 3 (B08, B09, B11, B12). B02–B04 helped design the configurations.

4. Input

4.1 The 8 inputs (one row per measurement)

#	Input	What it represents	Used?
x1	ln RMS	overall vibration energy	yes
x2	ln peak	the largest spike	yes
x3	kurtosis	impulsiveness, the impacts typical of bearing damage	yes
x4	ln band energy 1–5 kHz	mid-frequency vibration	yes
x5	ln band energy 5–20 kHz	high-frequency vibration, an early sign of bearing defects	yes
x6	crest factor	peak relative to RMS	yes
x7	shaft speed	the operating point	yes
x8	temperature residual	measured minus expected temperature	**no, always fed 0**

4.2 From sensor to network input

1. **Sensor:** one accelerometer (the rear one), converted to g. x1–x3 and x6 are computed at 42.67 kHz; the band energies come from the full-rate spectrum.
2. **Commissioning:** each bearing calibrates itself on its own first 100 operating measurements. Each feature is corrected for speed ($f = b0 + b1 \cdot \ln rpm$) and turned into a z-score against that period. Nothing is fitted across bearings.
3. **Log compression:** $x = \text{sign}(z) \cdot \ln(1 + \min(|z|, 10{,}000))$. This keeps large deviations ordered instead of cutting them off.
4. **INT8:** $x_q = \text{round}(x / 0.0725)$, where $0.0725 = \ln(10{,}001) / 127$.

5. The Network and Its 850 Parameters

Layer	Shape	Weights	Biases	Parameters	What they do
Hidden layer 1	8 → 32, ReLU	256	32	**288**	Each of the 32 units adds up the 8 inputs with its own weights (how much each feature counts, and in which direction) plus a bias (its switch-on level). Each unit detects one pattern, such as "energy and spikiness both up"
Hidden layer 2	32 → 16, ReLU	512	16	**528**	Combines the 32 patterns into 16 higher-level features
Deterioration head	16 → 1	16	1	**17**	Weighs the 16 features into one score (a logit). **Trained**
Health-index head	16 → 1	16	1	**17**	Would give the Motor Health Index. **Untrained and frozen**
Total		**800**	**50**	**850**	

Parameters that are actually active in the final model (seed 0):

- **Trained:** 833 of the 850. The health-index head's 17 stay at their initial values.
- **Temperature weights:** the 32 weights connecting x8 have decayed to about 0 (largest 7×10^{-14}), because x8 is always 0.
- **Inactive units:** 3 of the 32 first-layer units and 1 of the 16 second-layer units are switched off.
- **Near-zero weights:** in all, 132 weights are effectively zero. They are set to exactly 0 in the INT8 export.

Weight decay prunes the network in this way: it keeps the parameters it needs and shrinks the rest.

6. How It Trains

Setting	Value
Loss	binary cross-entropy on the deterioration score, counting **only labelled rows**. Unlabelled rows and the health-index head contribute nothing
Optimizer	Adam, learning rate 0.001, **weight decay 0.01**
Steps	2,000 steps of 64 rows (128,000 rows drawn; each labelled row is seen about 170 times)
Class balance	each batch is half healthy, half end-stage, drawn with replacement (the data are 700 : 35)
Temperature	x8 = 0 on every row, in training and in use
Seeds	5 (0–4); every result reproduces bit for bit
Time	about 2 minutes for all 1,225 Run 3 trainings, on 8 processes
Final training loss	0.019–0.022 (the 5 final networks)
Health-index head	excluded from the optimizer and checked bit-identical after training

How configuration E was chosen. Five configurations were compared under a rule fixed before training. The rule ranks, in order:

1. fewest healthy rows alarmed;
2. most end stages caught;
3. fewest alarm flickers;
4. lowest loss.

Alarm timing is not used, because the onset of damage is unknown. The first two criteria tied for every configuration, so the choice came down to stability. Configuration C (the same as E but with temperature) is statistically tied with E.

7. Output

Output	Definition	Status
Deterioration score	p = sigmoid(logit), from 0 to 1	trained
Alarm	on after **3 measurements in a row with p > 0.5**, then **latched** until a maintenance reset	trained, threshold fixed at 0.5
Motor Health Index	100 % × sigmoid(health output)	**disabled: never report it**

The score is a detector output with a threshold, not a calibrated probability of failure.

8. Results (held-out bearings)

8.1 Per bearing

These come from the 35 held-out networks: each bearing scored by the networks that never saw it, over 5 seeds. The anomaly detector is the label-free reference. It flags any departure from the bearing's own commissioning period.

Bearing	First alarm (measurement)	Minutes before the stop	Alarm flickers (unlatched)	Healthy rows alarmed	End stage caught	Anomaly detector, minutes before the stop
B02	M0867–M0868	80.0–80.3	1	0	yes	81.0
B03	M0561	16.5	1	0	yes	20.4
B04	M0445	208.6	1	0	yes	209.6
B08 (new)	M1216	214.1	1	0	yes	424.5
B09 (new)	M0226–M0230	644.0–645.5	5–7	0	yes	651.7
B11 (new)	M0712–M0713	337.9–338.1	1–21	0	yes	339.7
B12 (new)	never	–	0	0	**no**	40.8

"Minutes before the stop" is the time before the test was ended, not a verified warning time before failure.

8.2 Summary

Measure	Current model	Anomaly detector
Bearings whose end stage is caught	**6 of 7** (95 % interval 0.42–1.00)	7 of 7
New, unbiased bearings only	3 of 4	4 of 4
Healthy-row false alarms (latched)	**0 of 3,500**	0
First alarm	the same time or later on every bearing (2–19 measurements later on five, about 3.5 h later on B08, never on B12)	earlier or equal on every bearing
Alarm flickers (unlatched, mean per run)	**2.8**	1–14 per bearing
Highest score on a healthy row	0.54 (one B11 row, never 3 in a row)	–
Extreme scores (below 0.001 or above 0.999)	0.8 %	–

8.3 Row-level metrics, with a warning

These are computed on the labelled held-out rows: 100 healthy and 5 end-stage rows per bearing, × 7 bearings × 5 seeds = 3,675.

Accuracy	Precision	Recall	F1	TN / FP / FN / TP
0.9932	1.0000	0.8571	0.9231	3,500 / 0 / 25 / 150

> **Caution:** these numbers look better than the model is. The healthy rows are each bearing's own commissioning period, where the inputs are near 0, and the end-stage rows are extreme, so a rule that simply switches the alarm on right after commissioning scores 1.000. All 25 misses are B12. Judge the model by the per-bearing results in 8.1.

8.4 Why it misses B12 and never warns earlier

- **B12:**
  - Its end stage is weak on the rear sensor: the largest input is 2.5–3.2 standard deviations, against 18–51 on the other bearings.
  - Its stop was probably triggered by the front sensor, which the model doesn't use.
  - A vibration step inside its commissioning period weakens its inputs further.
- **Never earlier:** every end-stage label is an extreme, worn-out state, so the network learns to wait for extreme readings. The anomaly detector flags any departure from normal. Without labels for intermediate stages, a supervised network trained this way can't be expected to warn earlier.

9. Hardware

Item	Result
Format	INT8 weights, INT32 biases, 22-bit accumulators, 14 passes of the 8 × 8 array, both heads in one 16 × 2 layer
INT8 vs full precision (35 held-out networks)	13 of 45,015 decisions change (0.03 %); the first alarm is identical in all 35
INT8 vs full precision (5 final networks, in-sample)	39 of 45,015 decisions change (0.09 %)
RTL simulation	bit-exact (final seed 0, all 735 labelled rows, among 16 trained networks checked)
Speed	382 cycles = **1.78 µs** per measurement at 214.286 MHz; 1,642 cycles for 64 measurements
Board	**not yet run on the ZedBoard**
Exported files	`model/export/MTL_run3_v2/seed0–4/` (weights, biases and scales as hex, C header `mtl_run3.h`, `quant.json`, test vectors)

All five final networks (seeds 0–4) are exported. Seed 0 is the one checked on the full set of labelled rows in RTL. No single network has been frozen yet.

10. Limitations

- **Small sample:** only 7 bearings, and the bearing is the unit of evidence. No detection rate can be claimed.
- **Provisional labels:** the end-stage labels are proxies, the onset of damage is unknown, and the labels were not given a two-person sign-off (team decision).
- **Not better than the detector:** the model does not outperform the anomaly detector.
- **One sensor:** only the rear accelerometer is used, while two stops were probably triggered by the front one.
- **No temperature:** temperature is off. The data could not show that it helps (one sensor failed, and the bearings contradict each other).
- **Fixed threshold:** the 0.5 threshold is fixed, not tuned.
- **Uncovered regime:** the constant-speed regime (B14–B17) has no training bearing, and B17 is in the test set.
- **Simulation only:** all hardware results are from simulation.
- **Health index:** untrained.

11. Next Steps

1. **Combine the two detectors:** use the anomaly detector's per-bearing threshold as an input or a gate, with the MLP as a confirmer.
2. **Commissioning check:** add a validity check for the vibration commissioning period, which would catch B12's step.
3. **Sensor choice:** decide between the front sensor, the rear sensor, or both.
4. **More bearings:** add B10, B14 and B15 (outer race and constant speed).
5. **Threshold:** choose it inside the training folds instead of fixing it at 0.5.
6. **Final test:** freeze one model, test it once on B06, B07, B13 and B17, and report the result.
7. **Board:** run it on the ZedBoard, with the ARM code for features, commissioning and the alarm.

12. Files

What	Where
Model definition	`model/mtl_model.py`
Input pipeline	`model/paderborn.py` (features), `model/mtl_inputs.py` (commissioning, z-scores)
Labels	`model/labels.py`; table `model/results/supervised/run3/prep/label_table_run3.json`
Training	`model/train_deterioration_run3.py` → `model/results/supervised/run3/train/` (final networks in `models/FINAL/`)
Hardware export and checks	`model/run3_hardware.py` → `model/export/MTL_run3_v2/`, `model/results/supervised/run3/hardware/`
Full Run 3 report	`docs/Model/10_Training_run3.pdf`